TOAI 2026 Syllabus

Turkic Olympiad in Artificial Intelligence

Official Competition Syllabus

Version 1[1]

Introduction

The Turkic Olympiad in Artificial Intelligence (TOAI) is a premier global competition for high school students from Turkic-speaking nations, aiming to cultivate both a strong theoretical foundation and hands-on expertise in Artificial Intelligence. This syllabus outlines the topics contestants should master to excel in the competition.

Each year, the TOAI Scientific Committee updates the official syllabus to reflect the latest research findings and educational priorities.

Topic Classifications

The topics are categorized into three distinct sections, indicating the level and nature of knowledge contestants need:

1. **Theory (How it works)**: Contestants should understand core concepts and theoretical underpinnings—the "why" behind AI. This may involve studying textbooks, courses, and other resources to delve into the mechanics that power AI algorithms. Breadth should be prioritized over depth in covering all relevant topics.
2. **Practice (What it does, when to use it, and how to implement it)**: Contestants should develop practical skills necessary to implement AI methods in code. This includes knowing how to use library functions effectively, call the method on a particular data, and interpret outputs. *Example*: While a contestant need not fully dissect the internal workings of the Adam optimizer, they should be able to decide when and how to employ it.
3. **Both**: Certain topics require knowledge of both theoretical principles and practical application.

This structured approach ensures that contestants acquire the right balance of conceptual insight and hands-on proficiency across the diverse array of AI topics. Topics not listed are considered excluded from the syllabus. Questions can be directed to the scientific committee via the official website.

[1] Date: May 10, 2026

1 Foundational Skills & Classical Machine Learning

Topic	Subtopic	Category
Programming Fundamentals	Python Basics (Loops, Functions, etc.)	Practice
	NumPy and Pandas for Data Handling	Practice
	Matplotlib and Seaborn for Visualization	Practice
	Scikit-learn for ML	Practice
	PyTorch Basics	Practice
	Tensor (Multi-dimensional Array) Manipulation	Practice
	Training Models on CPU and GPU	Practice
Supervised Learning	Linear Regression	Both
	Logistic Regression	Both
	L1 & L2 Regularization	Both
	K-Nearest Neighbors (K-NN)	Both
	Decision Trees	Both
	Model Ensembles (Gradient Boosting, Bagging, Random Forest)	Practice
	Support Vector Machines (SVM)	Practice
Unsupervised Learning	K-Means Clustering	Both
	Principal Component Analysis (PCA)	Both
	t-SNE, UMAP	Practice
	DBSCAN, Hierarchical & Spectral Clustering	Practice
Data Science Fundamentals	Model Evaluation Metrics (Accuracy, Precision, Recall, etc.)	Both
	Underfitting, Overfitting	Both
	Hyperparameter Tuning	Theory
	Cross-Validation	Practice
	Confusion Matrix and ROC Curves	Practice
	Feature Engineering*	Both
	Data Processing**	Practice

* **Feature Engineering** involves transforming raw, potentially high-dimensional data, categorical data, time series, or ragged data into a compact set of informative features. Techniques involve sliding windows, pooling operations, one-hot encoding, statistical moment-based features (average, standard deviation), PCA and neural-network-based embeddings.

** **Data Processing** concerns the handling of missing data and irregular data, including in sequence modeling settings. Techniques involve basic imputation strategies (mean/median/forward-fill for sequences) and padding for variable-length sequences. Covered here are also normalization and standardization techniques, train/validation/test splitting strategies, basic data augmentation (flipping, cropping, noise addition), tokenization and vocabulary building for text and audio and patching for images.

2 Neural Networks & Deep Learning

Topic	Subtopic	Category
Neural Networks	Perceptron Basics	Both
	Gradient Descent	Both
	Backpropagation	Both
	Activation Functions (ReLU, Sigmoid, Tanh)	Both
	Loss Functions (MSE, MAE, Cross Entropy, etc.)	Both
Deep Learning	Multi-Layer Perceptrons (MLP)	Both
	Data Embeddings (text, image, audio)	Both
	Pooling Techniques (Max, Average)	Both
	Attention Mechanism	Both
	Transformers (theory needed only for text and image)	Both
	Autoencoders	Practice
	SGD, Mini-Batch Gradient Descent	Both
	Momentum Methods (Adam, AdamW)	Practice
	Convergence and Learning Rates	Practice
	Regularization: Dropout, Early Stopping, Weight Decay	Practice
	Weight Initialization	Practice
	Batch Normalization	Practice
	Model Finetuning (full and parameter-efficient)	Practice

3 Computer Vision

Topic	Subtopic	Category
Fundamentals	Convolutional Layers	Both
	Image Classification	Practice
	Object Detection (YOLO, SSD, DERT)	Practice
	Image Segmentation (U-Net)	Practice
	Pre-trained Vision Encoders (e.g. ResNet)	Practice
	Image Augmentation	Practice
	Generating Images with GANs	Practice
	Self-Supervised Learning for Vision	Practice
	Vision-text encoders (e.g. CLIP)	Practice
	Diffusion Models	Practice

4 Natural Language Processing & Audio

Topic	Subtopic	Category
NLP	Text Classification	Practice
	Pre-trained Text Encoders (e.g. BERT)	Both
	Language Modeling	Both
	Encoder-Decoder Models (e.g. for Machine Translation or Vision-Language Modeling)	Practice
	Pre-trained Language Models (open-source and API-based ones)	Practice
Audio Processing	Pre-trained Audio Encoders: HuBERT	Practice
	Audio Models: Qwen-Audio, Whisper, Voxtral	Practice

Note: The data can be text, tabular, image, audio, video, and time-series, and should be processed with the methods above.